

Economic growth

Economic growth is a key indicator of macroeconomic performance. Economic growth is an increase in an economy's output. The economic growth rate is the annual percentage change in output. For people to enjoy more goods and services, output has to increase by more than any growth in population. In such a case, GDP per head (per capita) would increase. For many years, it was assumed that poverty would be eradicated if countries managed to sustain economic growth. As a result, economic growth and economic development were seen as the same thing.

It was assumed that if economies grew they would also experience development. The increased availability of goods and services in an economy would lead to a 'trickle down' effect that would have an impact upon all, including the poor members of society, in terms of jobs and other economic benefits. In reality, however, economic growth does not result in a rise in the living standards and quality of life of everyone in an economy. It is also possible for a high proportion of people to achieve an improvement in their living standards and quality of life even if economic growth does not occur, for example, if there is a more equal distribution of income or a reduction in pollution. As a result, a wider definition of economic development is now accepted that is related to, but distinct from, economic growth. In other words, economic development is the process of improving people's economic well-being and quality of life.

Measurement of economic growth

Economic growth is measured in terms of changes in real GDP that is the country's output. The economic growth rate is the percentage change in real GDP from one time period to another, usually a year. Government statisticians first measure GDP in the prices operating at the time. They then take out the effects of changes in the price level to get real GDP.

Nominal GDP and real GDP

Nominal (or money) GDP is GDP measured in terms of the prices operating in the year in which output is produced. It is sometimes referred to as GDP at current prices and is a measure that has not been adjusted for changes in the price level.

Nominal GDP may give a misleading impression of how well a country is performing. This is because the value of nominal GDP may rise not because more goods and services are being produced but simply because prices have risen. For example, if 100 billion products are produced at an average price of \$5, GDP will be \$500 billion. If in the next year the same output of 100 billion products is produced but the average price rises to \$6, nominal GDP will rise to \$600 billion. So to get a truer picture of what is happening to output, economists convert nominal GDP into real GDP.

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To calculate real GDP, economists measure GDP at constant prices (the prices operating in a selected year). This ensures the effect of inflation that distorts nominal GDP is removed. For example, in 2016, a country's GDP is \$800 billion and the price index is 100. In 2017, nominal GDP is \$900 billion and the price index is 120. Using the formula to calculate real GDP:

$$\text{Real GDP} = \text{nominal GDP} \times \frac{\text{price index in base year}}{\text{price index in current year}}$$

$$\$900 \text{ billion} \times \frac{100}{120} = \$750 \text{ billion}$$

The price index used to convert nominal into real GDP is called the GDP deflator, which measures the prices of products produced, rather than consumed, in a country. So it includes the prices of capital goods as well as consumer products and includes the price of exports but excludes the price of imports.

Causes and consequences of economic growth

If there is spare capacity, output can increase as a result of an increase in aggregate demand. For instance, greater consumer confidence may lead to higher consumer expenditure and so an increase in aggregate demand. Similarly, an increase in government spending and a cut in taxes and the rate of interest may result in more factors of production being employed and output rising. Making more use of existing resources can be shown on a production possibility curve diagram, as shown in Figure 18.2.

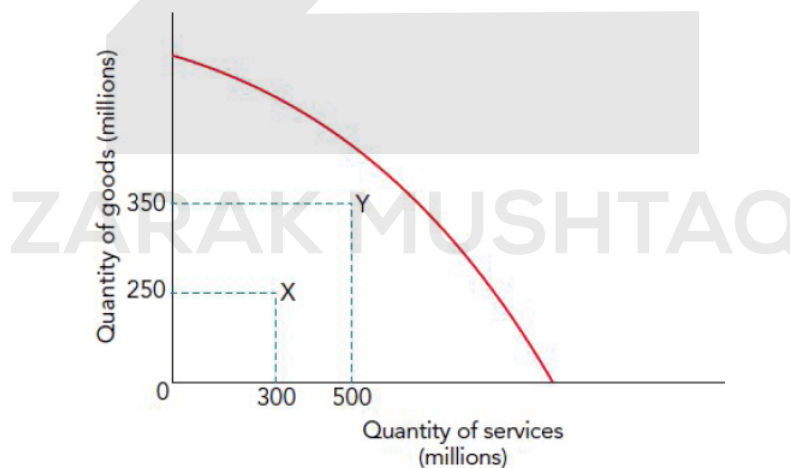


Figure 18.2: Making more use of existing resources

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The economy is initially producing at point X. Then the production point increases to point Y and more goods and services are produced. Economic growth resulting from higher aggregate demand can also be shown using an AD/AS diagram. Figure 18.3 shows an increase in aggregate demand in an economy with spare capacity. The increase in aggregate demand brings into use previously unemployed resources, and output (measured by real GDP) increases from Y to Y₁.

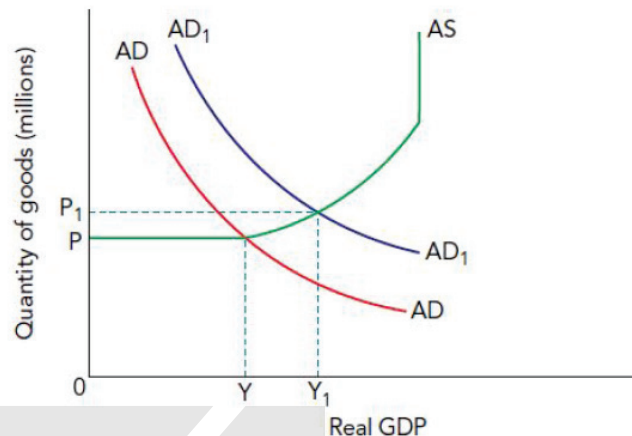


Figure 18.3: An increase in aggregate demand resulting in higher output

In order to achieve economic growth that can be sustained over time, it is necessary for productive capacity and aggregate supply to increase. There are two broad causes of such an increase: more resources or better quality resources.

Increase in the quantity of resources

The supply of labour and entrepreneurs could increase, after a period of time, as a result of a natural increase in population. More immediately, net immigration of people of working age is likely to lead to a greater availability of workers and entrepreneurs. The number of workers and entrepreneurs may also increase as a result of government policy measures. For example, a rise in the retirement age increases the supply of labour, and deregulation and privatization may promote entrepreneurship.

The supply of capital goods will increase if there is net investment, with firms buying more capital goods than are needed to replace those capital goods that are taken out of use. The quantity of land may increase by discoveries, for instance, of new oil fields and gold mines.

Increase in the quality of resources

Improvements in the quality of resources will increase the productivity of the factor inputs. The quality of labour and entrepreneurship can be improved through education and training and better healthcare. The quality of capital goods improves as technology advances. The use of fertilisers and irrigation and drainage schemes may improve the quality of land.

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Figure 18.4 shows productive capacity increasing using both a production possibility curve and an AD/AS diagram. Both diagrams illustrate an increase in the maximum output the economy is capable of producing. The first diagram shows that the production possibility curve has shifted to the right. The ability of the economy to make both consumer goods and capital goods has increased. It is now possible to produce a higher maximum combination of the two types of good. The second diagram also shows an increase in the economy's productive capacity. This time it is illustrated by a shift of the LRAS curve to the right.

For an increase in productive capacity to lead to higher output, the rise in productive capacity must be put to use. Figure 18.5 again uses two diagrams, this time to show both an increase in aggregate demand and an increase in aggregate supply. The first diagram shows an economy initially producing at point X inside the PPC. Then both productive capacity and aggregate demand increase and the economy's output rises to point Y. While production point Y represents a higher output than point X, both are inside a PPC and illustrate inefficiency. In both cases, the economy is not producing as much as it is capable of making. It is wasting resources. There may be, for instance, unemployed workers and unused machines. It would be possible to produce more of one type of good without having to give up any of the other type of good.

The second diagram also shows both productive capacity and output has increased. The shift to the right in the AD and LRAS curves raises the country's output from Y to Y₁. Again, however, the economy is not operating at maximum

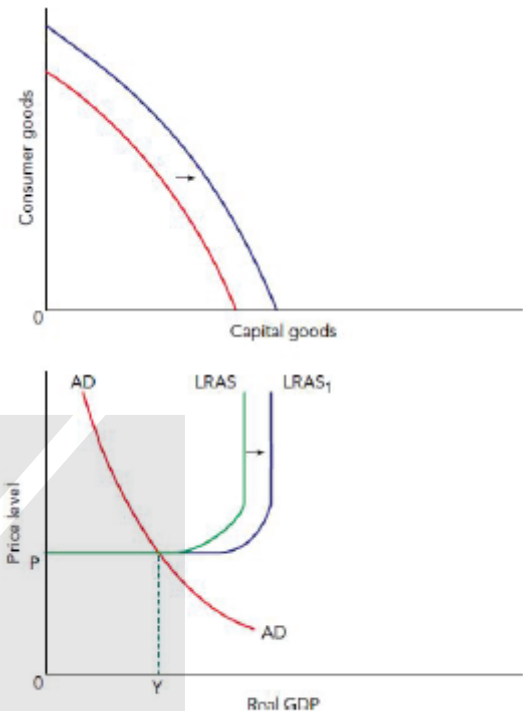


Figure 18.4: Increase in productive capacity shown on a PPC diagram (top) and AD/AS diagram.

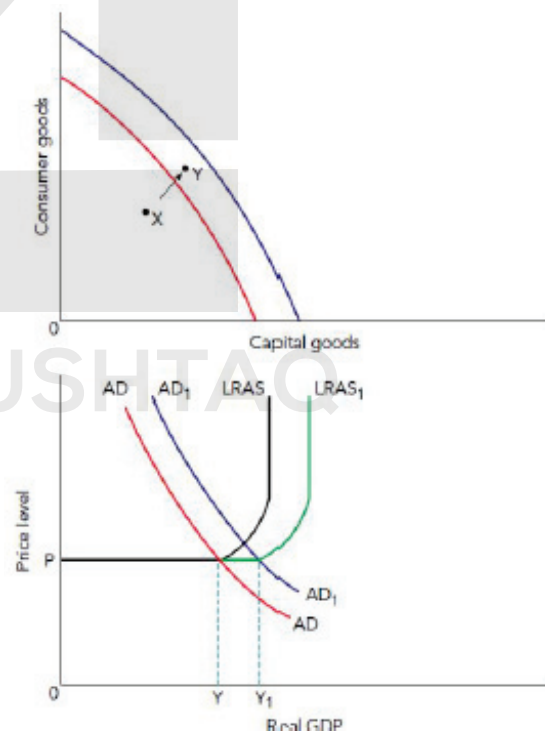


Figure 18.5: Economic growth resulting from a combination of higher aggregate demand and aggregate supply

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capacity either when it is making Y or Y_1 . Maximum output would be achieved if the AD curve cuts the LRAS curve on the vertical part of the LRAS curve.

Economic growth can be negative as it is possible that a country's real GDP may fall. A decline in real GDP over two consecutive quarters (six months) or more is called a recession. A country's output may decline as a result of a decrease in aggregate demand or a decrease in aggregate supply.

Economic growth in low-income countries

The main obstacle to increases in the quantity and quality of resources in some low-income countries is the opportunity cost of allocating resources away from their current use. For example, allocating more resources to education may mean that fewer resources can be allocated, for instance, to healthcare. Producing more capital goods may mean sacrificing consumer goods in the short run, which would lead to a decrease in living standards. Research and development that can result in advances in technology also use resources that have alternative uses.

Economic growth in China and India

China and India have witnessed particularly rapid economic growth. China has become one of the two largest economies in the world. It accounted for 30% of global economic growth between 2012 and 2019. China's increase in output has been driven by increases in investment and exports. The high rates of saving in China have freed up some resources for investment and the relatively low value of the Chinese currency, the renminbi, has contributed to the rise in demand for Chinese exports.

India's economic growth rate has been boosted by an increase in the size of the labour force and advances in information technology. Between 2012 and 2019, India accounted for 16% of global economic growth.

The costs and benefits of economic growth

Costs

As explained above, if an economy is operating at full capacity, there will be an opportunity cost involved in achieving economic growth. To produce more capital goods, in order to increase the country's productive capacity, some resources will have to be moved from producing consumer goods to producing capital goods. The consequence is that the current consumption of goods and services will have to be reduced. However, this will only be a short-run cost since, in the long run, increased investment will increase the output of both capital goods and consumer goods and services.

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Some other potential costs may exist in both the short run and the long run, including increased stress and anxiety. A growing economy is a dynamic (changing) economy that undergoes structural changes, with some industries expanding and some declining. Workers may have to learn new skills and may have to change their occupation and/or where they live. Some workers may find such changes difficult to cope with. Economic growth may also be accompanied by increased working hours and pressure to come up with new ideas and improvements. There is a major difference in the number of hours people work in different countries. In 2018, for instance, the average annual hours worked per person in Mexico was 2148 whereas it was only 1363 in Germany.

In addition, economic growth may be accompanied by the depletion of natural resources and damage to the environment. Higher output may, for example, involve firms using more oil, depleting fish stocks, building on areas of natural beauty and creating more pollution.

Benefits

The main benefit of economic growth is the increase in goods and services that become available for the country's citizens to enjoy. This can raise people's living standards. Economic growth makes it easier to help the poor. Higher incomes and more spending increase tax revenue and some of this extra revenue may be given to the poor in the form of higher benefits, better housing, better education and better healthcare.

Without any increase in output and income, a government may have to raise the tax rates on higher income groups, and so reduce their living standards, in order to help the poor. Economic growth may also be accompanied by a rise in employment. A rise in real GDP caused by higher aggregate demand is likely to create extra jobs. An increase in aggregate supply may make a country's products more internationally competitive and so may generate more jobs.

A stable rate of economic growth tends to increase business and consumer confidence, which encourages investment. Economic growth in itself can help encourage further economic growth. In addition, economic growth may increase a country's international prestige and power. For example, China's rapid economic growth since the early 1990s has increased its status as an economic and political power. Some economists in high-income countries debate whether the benefits of economic growth outweigh the costs. For those in low-income countries, however, economic growth is seen as essential to bring people out of poverty.